MASTER TECHNOLOGY & LEARNING PLAN

International Student Transition & Career Intelligence Platform

A long-term backend and data engineering roadmap tracking tech stacks, core targets, and engineering knowledge modules.

PROJECT CORE FOCUS

This roadmap presents the comprehensive development lifecycle of the **DE-Transit Platform**. It maps how the codebase systematically scales from foundational SQL scripts into automated analytical structures, bulk ETL processing pipelines, and production web services. Every phase is explicitly contextualized to handle the rules of German bureaucracy (Anmeldung, Visas), student requirements, and corporate working student (*Werkstudent*) application pipelines.

PHASE 1 Python + Relational SQL Foundations

Core Target: Construct the data structure layer modeling four relational modules: Students (profiles, visa paths), Document Types (lookup rules for Anmeldung/Insurance), Companies (employer registry), and Applications (the master many-to-many process tracking junction).

🛠️ WHAT WE WILL USE:

- **Python 3.12 & Pip:** Standard virtual environments (venv) and modular folder isolation.
- **PostgreSQL Engine:** Database servers hosted locally for relational integrity.
- **Psycopg2 Driver:** The basic pipeline broker to send commands from Python to SQL.
- **SQL Commands:** Direct writing of `CREATE TABLE`, `INSERT`, `SELECT` with `JOIN`, `GROUP BY`, and constraints.

🧠 WHAT YOU WILL LEARN:

- **Relational Modeling:** Structuring one-to-many paths and building complex junction data matrices.
- **Manual Query Control:** Merging datasets using explicit filters and primary/foreign key connections.
- **Transaction Lifecycles:** Mastering safe state commits and rolling back failed database changes.

PHASE 2 Refactoring + Professionalization

Core Target: Modernize the foundational core by transforming loose procedural scripts into encapsulated, dry, safe application blocks driven by a polished terminal interface.

🛠️ WHAT WE WILL USE:

- **Python Abstraction Layers:** Functional decomposition and decoupled business logic layers.
- **Regular Expressions (Re):** Text search patterns for email, date, and input validation.
- **Python Exception Tree:** Targeted interception of `DatabaseError` or environment flaws.
- **CLI Navigation Loops:** Modular pagination formatting and clean table layouts.

🧠 WHAT YOU WILL LEARN:

- **Defensive Programming:** Building rigorous input guards to trap user errors before database transmission.
- **Separation of Concerns:** Isolating operational database queries away from raw terminal presentation code.
- **Code Maintainability:** Eliminating hardcoded parameters and designing structured, scalable functions.

PHASE 3 Advanced SQL + Analytics Engine

Core Target: Convert your storage layer into an intelligence reporter tracking student visa backlogs, interview funnel conversions, and German market salary analytics.

🛠️ WHAT WE WILL USE:

- **Common Table Expressions (CTEs):** Writing highly readable, multi-tiered data isolation scopes.
- **Window Functions:** Running localized analytics rankings over partitioned rows.
- **Database Views:** Saving advanced query logic behind permanent visual structures.
- **PostgreSQL Indexing:** Deploying B-Tree and hash indices on frequently searched keys.

🧠 WHAT YOU WILL LEARN:

- **Analytical Thinking:** Writing query blocks to synthesize raw metrics into structured reporting dashboards.
- **Query Optimization:** Analyzing execution plans to profile bottlenecks.
- **Conditional Pipelines:** Reshaping query layouts using nested database logic.

PHASE 4 Pandas + Data Engineering Workflows

Core Target: Create data ingestion engines to process external files (like public university lists or city office datasets), run cleanings, and synchronize state.

🛠️ WHAT WE WILL USE:

- **Pandas Library:** High-performance DataFrames built for structural matrix tracking.
- **CSV/JSON Engines:** Input/output modules handling raw transactional files.
- **SQLAlchemy Engine:** Streamlining high-speed block inserts to PostgreSQL.
- **Data Transformation Pipelines:** Modular mapping scripts executing row joins.

🧠 WHAT YOU WILL LEARN:

- **ETL Architecture:** Constructing clean pipelines to extract, transform, and load data.
- **Data Vectorization:** Leveraging Pandas vector mechanics to replace slow iterative loops.
- **Data Scrubbing Strategy:** Detecting null profiles and handling outlier anomalies.

PHASE 5 FastAPI Backend Service

Core Target: Completely migrate the terminal command execution layout into a high-speed, modern, production-grade REST API server ecosystem.

🛠️ WHAT WE WILL USE:

- **FastAPI Framework:** High-performance ASGI service networks built on Starlette.
- **Pydantic Core:** Strict data payload serialization and structural modeling.
- **Uvicorn Server:** Lightning-fast ASGI production execution pathways.
- **HTTP Protocol:** REST verbs (`GET`, `POST`, `PUT`, `PATCH`, `DELETE`) and status returns.

🧠 WHAT YOU WILL LEARN:

- **API Architecture:** Designing cleanly segregated, decoupled backend systems.
- **Type Serialization:** Enforcing compile-time schema safety validations automatically.
- **Client-Server Communication:** Managing web payloads, query params, and responses.

PHASE 6 Containerization + DevOps Foundations

🛠️ WHAT WE WILL USE:

- **Docker Configurations:** Standardizing code isolation environments via `Dockerfiles`.
- **Docker Compose:** Synchronizing network channels between app and database.
- **Environment Vaulting:** Decoupling production configs from source code files.

🧠 WHAT YOU WILL LEARN:

- **Infrastructure as Code:** Creating predictable, consistent builds across environments.
- **DevOps Awareness:** Orchestrating separate containers onto unified networks.

🎯 THE CORE ENGINEERING OUTCOME

By tracking through this technology matrix step-by-step, you are not just building an application; you are establishing complete backend fluency. You will demonstrate a profound understanding of database designs, optimized data reporting pipelines, robust text parsers, and API architectures. This provides clear, invaluable discussion points for technical interviews with engineering teams in Germany.